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Lossless Compression of Modern Astronomical Data Using a Novel Learned Predictor

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Abstract

The rapid growth of science-ready data volumes generated by modern ground-based astronomical observatories poses significant challenges for data storage and transmission. The de facto standard for storing these data is FITS, which incorporates a limited set of built-in compression methods. In this work, we present a novel lossless compression approach for astronomical images based on a weighted linear predictor combined with a CCSDS 123.0-B-2 mapper and a contextual binary arithmetic encoder. Three predictor configurations are evaluated, ranging from handcrafted designs to an optimized, dataset-adaptive model. Experimental results show consistent improvements over state-of-the-art lossless compressors. The optimized predictor achieves the best performance, improving compression by up to 0.471 bits per pixel (6.40%) compared to the best-performing FITS compressor, FPACK Hcompress. When applied to large-scale instruments such as the VLT, this corresponds to a reduction of approximately 0.25 TB per day (4.48%), yielding annual storage savings exceeding 90 TB. The implementation is publicly available at https://github.com/xavifeme00/Astronomy-Compressor.

Unified Astronomy Thesaurus concepts: Astronomy software (1855); Astronomy data reduction (1861); Astronomy data analysis (1858)

1. Introduction

The deployment of new ground-based observatories (Burgett et al. 2024), along with the modernization of existing ones (Hayoz et al. 2025), is leading to a significant increase in the amount of astronomical data (Khalil et al. 2019). The continuous improvement of sensors, cameras, and observational techniques causes the volume of information to grow rapidly, demanding more efficient solutions for its management and scientific exploitation (Denker et al. 2018; Poudel et al. 2022). Several authors have pointed out that astronomy has now entered the era of data-intensive science, where the scale, complexity, and diversity of datasets require not only traditional observational techniques but also advanced computational and data management approaches (Zhang & Zhao 2015; Rahman et al. 2019; Zhang & Luo 2025).

As shown in Figure 1, the next generation of surveys and observatories is projected to generate petabytes of astronomical images per year, highlighting the critical importance of efficient data compression strategies (VIRCAM VISTA User Manual 2009; OSIRIS User Manual 2013; Pan-STARRS 2016; ERIS User Manual 2022, 2025; LSST Technology 2025). Despite these increasing data volumes, most observatories continue to store their data in the FITS format.

The Flexible Image Transport System (FITS; Hanisch et al. 2001) has served as the standard file format for storing and exchanging astronomical data from both ground-based and space-borne telescopes. FITS was originally developed in the late 1970s by astronomers and data specialists from NASA. Shortly after its introduction, the International Astronomical Union formally approved it as the standard format for the global astronomy community, thereby endorsing its use across observatories and institutions worldwide. A FITS file consists of two primary components: the header and the data unit. The header contains human-readable ASCII metadata that describes the scientific and technical properties of the observation: the instrument, exposure times, wavelength bands, coordinate systems, and other relevant parameters. The data unit stores the actual image, table, or spectrum. This structure ensures that FITS files are both self-describing and portable across platforms and software generations. FITS supports the FPACK suite of compression algorithms (Pence et al. 2011), also developed by NASA and specifically tailored for astronomical data. The suite includes both lossless and lossy modes, enabling FITS to handle large datasets with reasonable storage and transmission efficiency. Nevertheless,

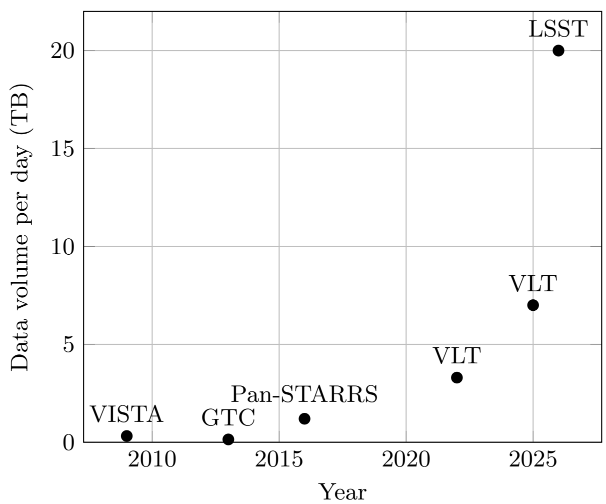


Figure 1. Evolution of the data volume produced per day by major astronomical surveys, assuming 10 hr of observation. For observatories equipped with multiple instruments, the following were selected for consistency: OSIRIS for the Gran Telescopio Canarias (GTC) and ERIS for the Very Large Telescope (VLT). The two data points corresponding to the VLT reflect successive upgrades.

the compression techniques currently provided by FPACK are not fully optimal in exploiting the spatial redundancy present in astronomical images, with Hcompress being the only algorithm in the suite that partially addresses this characteristic through the use of the Haar wavelet transform. To meet this issue, several authors have focused on developing more efficient compression methods for astronomical data. Recent efforts include neural-network-based lossless compression techniques (Williamson et al. 2025), lossy compression techniques Maireles-González et al. (2024) and Dodson et al. (2025), and the AstroCompress benchmark, which provides standardized datasets to evaluate and compare compression approaches (Truong et al. 2025).

In Maireles-González et al. (2023), we presented a study highlighting the rapid growth in data generation from next-generation ground-based telescopes compared to their predecessors, along with the lack of progress in the compression techniques employed. That paper illustrates a clear need for improving compression efficiency, given the dramatic increase in data volumes. In that work, we introduced a novel lossless compression algorithm that outperformed the conventional FITS coders and other widely used schemes, achieving a significant reduction in storage requirements over the best FPACK configuration. The study emphasized that the FITS standard, which had remained largely unchanged for decades, could and should be enhanced to better address the needs of modern astronomical data handling.

Today, telescopes that were once part of future projections have become operational realities. For instance, the Large Synoptic Survey Telescope (LSST) at the Vera C. Rubin Observatory (Ivezić et al. 2019) has already completed its first data preview (DP1) while still under construction, generating approximately 3.5 TB of data over a limited period of seven weeks in late 2024 (Vera C. Rubin Observatory Team Observatory Team 2026), which is accessible to authorized users via the Rubin Science Platform. Once fully operational, LSST is expected to produce around 20 TB of raw data per day (Rubin Observatory 2026), accumulating tens of petabytes over its lifetime.

To address this challenge, we present a novel *lossless* compression algorithm. Our approach outperforms all previously published methods, including the FPACK suite, and highlights the pressing need to develop and integrate new algorithms into the FITS standard. In parallel, we have expanded and diversified the evaluation dataset to enhance the robustness and generality of our assessment, incorporating data from more recent astronomical instruments. These advances are crucial to ensure that astronomical data management remains sustainable in the face of the unprecedented data volumes that have become a present reality rather than a distant projection.

This paper is organized as follows. Section 2 describes the astronomical datasets considered in this work. Section 3 details the proposed compression method. Section 4 presents and discusses the experimental results. Section 5 concludes the paper.

2. Astronomical Data

The dataset employed in this study extends the collection introduced in our previous work (Maireles-González et al. 2023), incorporating additional observations from several major ground-based telescopes currently in operation. The original dataset comprised images obtained from the Isaac Newton Telescope (INT), Jacobus Kapteyn Telescope (JKT), Las Cumbres Observatory (LCO), Telescopi Joan Oró (TJO), and the William Herschel Telescope (WHT). Most of these facilities are located at the Observatorio del Roque de los Muchachos in La Palma, Canary Islands (Spain). This initial compilation offered a broad variety of stellar fields differing in image size, resolution, and photometric depth. Further details on the original dataset can be found in our previous work.

To enhance the dataset's diversity and representativeness, we incorporated new observations from two additional facilities: the Gran Telescopio Canarias (GTC) (Instituto de Astrofísica de Canarias (IAC) 2007) and the Very Large Telescope (VLT) in northern Chile (European Southern Observatory (ESO) 1998).

Data from the GTC were acquired using the OSIRIS instrument. Four distinct sky regions were selected with ten images obtained for each target,[3] resulting in forty new samples. The images were retrieved from the GTC Science

[3] 2024_space_PT5, DR3Gaia210752-161131, GRB 240919A, and WISE0523-0153.

Table 1
Characteristics of Each Dataset Used

	TRAIN			TEST		
Telescope	Images	Dimensions	Average Entropy (bpp)	Images	Dimensions	Average Entropy (bpp)
INT	10	4200 × 2154	6.13	30	4200 × 2154	5.66
JKT	10	[851, 2148] × [941, 2148]	7.54	32	[1000, 2148] × [1000, 2148]	7.15
LCO	10	2112 × 3136	5.74	53 18 3	2112 × 3136 2048 × 2080 510 × 765	5.78
TJO	10	4096 × 4096	5.50	7 4	4096 × 4096 4108 × 4096	6.36
WHT	10	[520, 4200] × [550, 2148]	6.18	30	[2501, 2981] × [2148, 2154]	8.00
GTC	10	2056 × 2073	11.31	30	2056 × 2073	9.97
VLT	10	2046 × 2048	12.83	10 10 10	2046 × 2048 1024 × 2048 512 × 2048	12.83

Note. Entropy results correspond to zero-order entropy, measured in bits per pixel (the number of bits required to represent a single pixel).

Data Archive. (Solano & Gutiérrez 2009). Similarly, data from the VLT were obtained using the EIRIS instrument, targeting four astronomical objects.[4] Ten images were selected for each object, yielding an additional forty samples. These data were retrieved from the ESO Science Archive Facility (Romaniello 2022). All image links can be found at gici.uab.cat (the full URL is available in the digital version).

Since the proposed compression model includes an initial training stage, the dataset was divided into disjoint training and testing subsets. To construct the training and testing sets for the proposed approach, we selected the first 10 images from each dataset for training, while the remaining images were reserved for evaluation. The training set was used exclusively to optimize the algorithm and adjust its parameters, whereas the test set was employed to evaluate performance and to report the results for all compared methods, including the proposed approach. The dataset split is detailed in the GitHub repository.

Table 1 summarizes the final datasets, detailing, for each telescope, the total number of images, their typical spatial dimensions, and the corresponding average entropy values. In addition, Figure 2 provides a visual overview of the datasets, illustrating the diversity in resolution, field of view, and stellar density across the different telescopes.

3. Proposed Approach

3.1. Related Work

To evaluate the performance of lossless compression algorithms in the field of astronomy, several algorithms were tested. Among them, FPACK—the most widely adopted solution in astronomical data processing—includes three algorithms specifically optimized for scientific imagery: (i) Rice compression (Rice et al. 1993) is the most commonly employed due to its simplicity, speed and suitability for integer-valued astronomical data; (ii) GZIP (Deutsch & Gailly 1996), based on the DEFLATE algorithm, is also supported, offering a balance between compression ratio and computational cost; and (iii) Hcompress (White et al. 1992) uses a wavelet-like transform (the H-transform) and is capable of both lossless and lossy compression. Thanks to the transform, this algorithm can effectively exploit spatial redundancy.

Beyond FPACK, other compression schemes have also been investigated in the context of astronomical data management. A prominent family of approaches derives from the JPEG standards: (i) JPEG 2000 (Taubman & Marcellin 2012), a compression standard based on a discrete wavelet transform that supports both lossy and lossless modes, making it especially well suited for large-scale image archives—in particular, the reversible standard 5/3 wavelet and the Haar transform implementation described in Maireles-González et al. (2023); (ii) JPEG-LS (Weinberger et al. 2000), based on predictive coding and context modeling, offering competitive lossless compression ratios with very low computational complexity; and (iii) JPEG XL (Alakuijala et al. 2019), representing a modern, open-source successor that combines efficiency with advanced features such as multi-resolution storage and improved entropy coding.

Moreover, we evaluate algorithms specifically tailored for space data. Fully Adaptive Prediction Error Coder (FAPEC) (de Mora et al. 2010), initially developed for the ESA GAIA

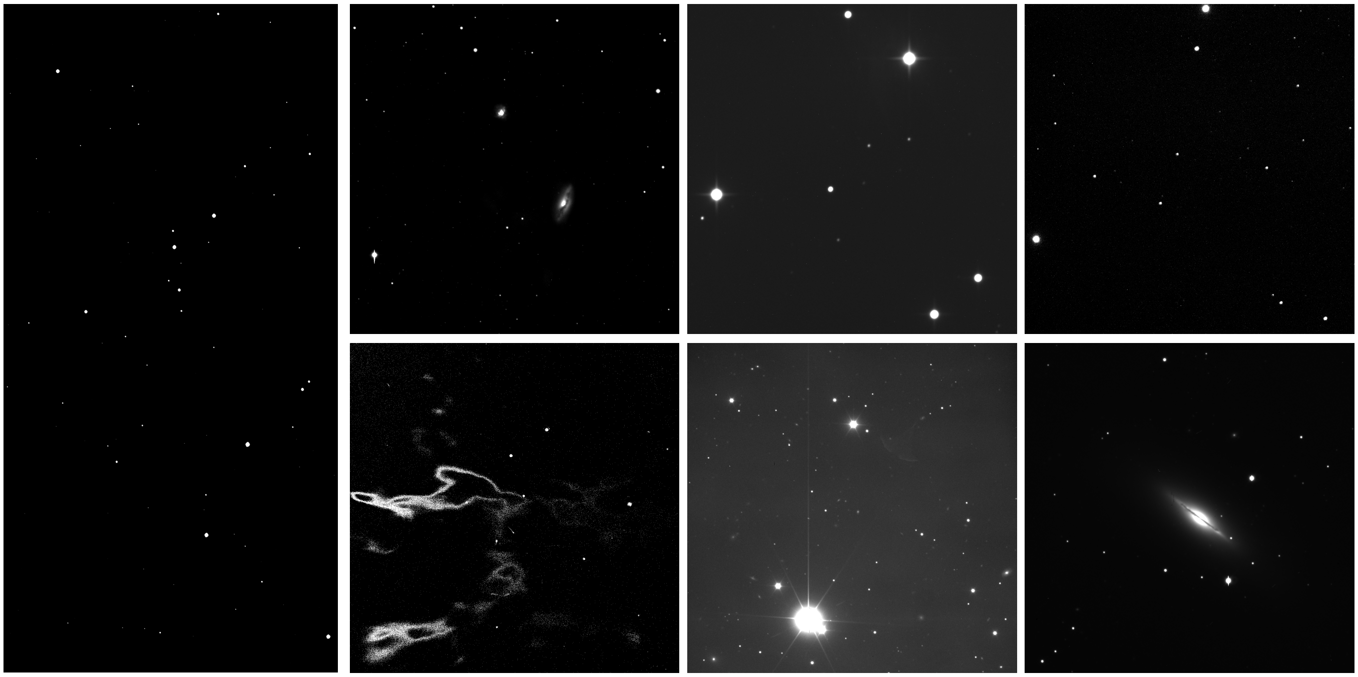
[4] V960 MON, CI TAU, ASAS J112755-6625.9, and Sagittarius A

(a) Telescope = INT
Entropy = 3.74

(b) Telescope = JKT
Entropy = 10.37

(c) Telescope = TJO
Entropy = 5.32

(d) Telescope = LCO
Entropy = 6.20



(e) Telescope = VLT
Entropy = 12.63

(f) Telescope = GTC
Entropy = 8.34

(g) Telescope = WHT
Entropy = 7.31

Figure 2. Crops of 1600 × 1600 pixels and 3200 × 1600 from different regions of the sky and different telescopes. Each image has been adjusted for visualization purposes; the brightness and contrast do not reflect the original flux levels. The entropy of each image is measured in bpp.

mission, is a robust and adaptive lossless compressor designed to handle data with outliers efficiently, achieving competitive ratios while maintaining low complexity. Another important standard is CCSDS 123.0-B-2 (Hernandez-Cabronero et al. 2021), developed by the Consultative Committee for Space Data Systems (CCSDS). This standard defines a predictive lossless compression algorithm for multispectral and hyperspectral images, optimized for use in remote sensing mission data systems.

Finally, other general purpose compressors are assessed. Lempel-Ziv-Markov chain algorithm (LZMA) (Pavlov 2001) and Zstandard (ZSTD) Collet (2016) are well-known for their excellent compression ratios and flexible trade-offs between speed and efficiency. HEVC (Sullivan et al. 2012), though primarily intended for video data, can be adapted for large 2D astronomical images due to its block-based prediction and transform coding. Bzip2 (J. Seward 1996), while slower, provides strong lossless compression and serves as a benchmark in comparative analyses.

3.2. Proposed Technique

We propose a novel lossless compression algorithm composed of three main components: a weighted pre-trained linear predictor, the CCSDS 123.0-B-2 mapper (Hernandez-Cabronero et al. 2021), and the contextual binary arithmetic coder described in Bartrina-Rapesta et al. (2017). Figure 3 depicts the three components. Among these, the predictor represents the primary contribution of this work.

The objective of the proposed algorithm is to train a predictor matrix for each dataset, which is convolved over the entire image to generate an estimated value for each pixel based on its local neighborhood. The prediction residuals, defined as the difference between the actual pixel values and the predicted values, are expected to exhibit lower entropy than the original image and, consequently, to be more compressible. The predictor weights are subsequently optimized to minimize the average entropy across the dataset. Implementing this approach poses certain challenges, which we address by providing a detailed mathematical description of the algorithm. These equations precisely define the method and ensure full reproducibility of the compression process.

Let $\mathcal{D} = \{X^{(1)},\ldots,X^{(N)}\}$ denote a dataset of N grayscale 16-bit images acquired by the same sensor on a given telescope. For a generic image $X^{(n)} \in \mathcal{D}$, with $X^{(n)} \in \mathbb{N}^{H\times W}$, we denote by $x_{ij}^{(n)}$ the pixel at row i and column j. For each pixel location (i, j), we define a local image patch $P_{ij}^{(n)} \in \mathbb{N}^{K_h\times K_w}$ ending at

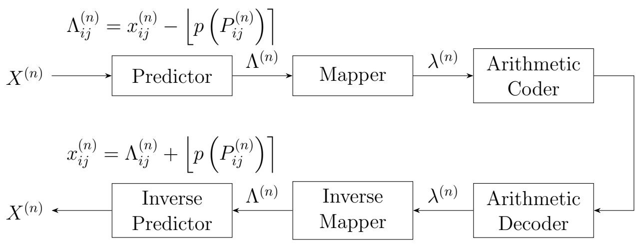


Figure 3. Block diagram of the proposed encoding and decoding technique.

row i and column $j + o_w$, where $o_w \geqslant 0$ is a horizontal offset that shifts the patch to the right:

$$P_{ij}^{(n)} = X^{(n)}[i - K_h + 1: i,\ \ j - K_w + 1 + o_w: j + o_w], \quad (1)$$

where rows range from $i - K_h + 1$ to i (inclusive) and columns from $j - K_w + 1 + o_w$ to $j + o_w$ (inclusive). We define the predictor function p: $\mathbb{N}^{K_h \times K_w} \longrightarrow \mathbb{R}$, which estimates the pixel $x_{ij}^{(n)}$ as a weighted sum of the patch values

$$p(P_{ij}^{(n)}) = \sum_{k=0}^{K_h-1} \sum_{l=0}^{K_w-1} P_{ij}^{(n)}[k, l] \cdot L[k, l] = \langle P_{ij}^{(n)}, L \rangle, \quad (2)$$

where $\langle \cdot , \cdot \rangle$ denotes the inner product and $L \in [0, 1]^{K_h \times K_w}$ is a weight matrix with real values. The image is processed using a line-by-line scanning order. To guarantee a valid and reproducible decompression process, the predictor is constrained to be causal: all weights associated with the current pixel (i, j) and with the pixels located to the right of (i, j) in the same row -that is, weights (i, j), $(i + 1, j)$, …, $(i + o_w, j)$- are set to zero, ensuring that only previously encoded pixels contribute to the prediction. Finally, during compression, the prediction residual $\Lambda_{ij}^{(n)} \in \mathbb{Z}$ at pixel (i, j) is defined as the difference between the true pixel value and the predicted value, rounded[5] to the nearest integer to match the discrete nature of pixel values

$$\Lambda_{ij}^{(n)} = x_{ij}^{(n)} - \lfloor p(P_{ij}^{(n)}) \rceil. \quad (3)$$

During decompression, the pixel value $X_{ij}^{(n)}$ is reconstructed using the residual $\Lambda_{ij}^{(n)}$, derived from the inverse mapper function, together with the corresponding patch composed of previously decompressed pixels

$$x_{ij}^{(n)} = \Lambda_{ij}^{(n)} + \lfloor p(P_{ij}^{(n)}) \rceil. \quad (4)$$

The complete end-to-end processing pipeline is illustrated in Figure 3. Additionally, Figure 4 provides a visual representation of each matrix involved in the process.

The objective of the training process is to determine, for each dataset $\mathcal{D}$, the weight matrix L that minimizes the final compressed dataset size. Since the arithmetic coder relies on contextual binary entropy coding using the HVD context model strategy (see Bartrina-Rapesta et al. (2017)), the training

[5] We use $\lfloor \cdot \rceil$ to denote rounding to the nearest integer.

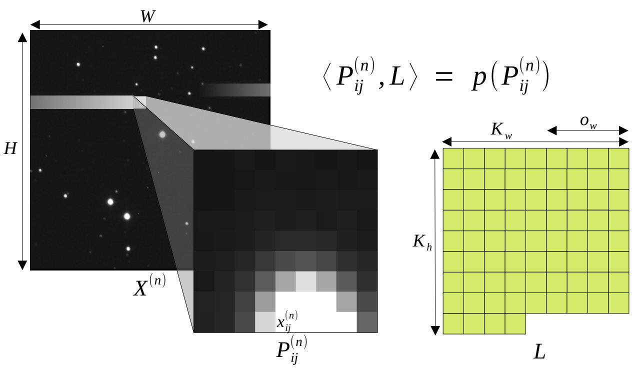


Figure 4. Visual representation of the image $X^{(n)}$, patch $P_{ij}^{(n)}$ and weight matrix L.

procedure focuses on minimizing the average entropy of the prediction residuals $\Lambda_{ij}^{(n)}$ across all images in the dataset, following the same entropy modeling scheme employed by arithmetic coding. The optimization is performed under the constraint that the elements of the weight matrix L add up to one.

Formally, the optimization problem is defined as

$$L^* = \arg\min_{L} \frac{1}{N} \sum_{n=1}^{N} H_{\mathrm{HVD}}(\Lambda^{(n)}(L))$$
$$\text{s.t.} \sum_{k=0}^{K_h-1} \sum_{l=0}^{K_w-1} L[k, l] = 1, \quad (5)$$

where $H_{HVD}(\Lambda^{(n)}(L))$ denotes the contextual binary entropy of the residuals using the L matrix.

3.3. Training

For the optimization problem, we adopt batch gradient descent. The loss function is defined as the average contextual entropy introduced in the optimization formulation:

$$\mathcal{L}(L) = \frac{1}{N} \sum_{n=1}^{N} H_{\mathrm{HVD}}(\Lambda^{(n)}(L)). \quad (6)$$

Accordingly, the weights L are learned by iteratively updating them using the gradient of the entropy. However, the gradient cannot be computed analytically from Equation (6) due to the presence of the binning operation in the entropy calculation and due to the non-differentiable round operator in Equation (3). Several authors address this issue using surrogate gradient techniques, most notably the soft binning function (Ustinova & Lempitsky 2016) for binning and the Straight-Through Estimator (STE; Huh et al. 2023) for rounding. In our case, we opted for numerical gradient estimation due to the small number of parameters and to avoid introducing heuristic gradient approximations. The numerical gradient is computed by perturbing each weight with a small increment ε. The resulting change in the loss function allows us to estimate the gradient along each dimension by evaluating the average entropy before and after the perturbation and determining the direction of steepest descent:

$$\frac{\partial \mathcal{L}}{\partial L[k, l]} \approx \frac{\mathcal{L}(\Pi(L + \varepsilon E_{kl})) - \mathcal{L}(L)}{\varepsilon}, \quad (7)$$

where E_{kl} is a zero matrix except for a 1 at position (k, l) and $\Pi(\cdot)$ denotes the normalization operator enforcing $\sum_{kl} L[k, l] = 1$. Once the gradients for all dimensions have been estimated, the weights are updated using a learning rate α:

$$L[k, l] \leftarrow \Pi\left(L[k, l] - \alpha \frac{\partial \mathcal{L}}{\partial L[k, l]}\right). \quad (8)$$

Before training, we define two handcrafted predictor weight matrices, each designed to capture different spatial characteristics of astronomical images. The first predictor, denoted as 2×2, considers only the three nearest causal neighbors of the current pixel—the pixel above, the pixel to the left, and the upper-left diagonal—assigning equal weights of $1/3$ to these neighbors and zero elsewhere. This predictor captures fine local details, allowing it to adapt well to edges and sharp intensity transitions. However, its limited spatial context makes it more sensitive to noise and local fluctuations. In contrast, the second predictor exploits a full 9×9 neighborhood with an offset $o_w = 4$, where all causal positions are uniformly weighted. By incorporating a substantially larger spatial context, this predictor produces smoother estimates that tend toward the local average, reducing prediction variance and improving robustness to noise, though at the cost of potentially smoothing fine details and rapid transitions. To complement these two handcrafted predictors, we train a third predictor with optimized weights, using a matrix with dimensions of $K_h = K_w = 9$ and an offset $o_w = 4$. This refined predictor is designed to adapt to the specific characteristics of each dataset, thereby improving compression performance. Since the optimization problem is inherently non-convex, the choice of the initial weight configuration is critical, as the learning process may converge to local minima. The weight matrix is therefore initialized using the 9×9 handcrafted predictor, as preliminary results show that it achieves better performance than the 2×2 configuration. During training, the weights are iteratively adjusted to determine which spatial contributions should be emphasized or attenuated. The predictor considers a maximum spatial extent of up to four pixels to the left, eight pixels above, and four pixels to the right of the current pixel.

The predictor is trained for 60 epochs with a $\alpha = \varepsilon = 10^{-3}$, aiming to converge toward an optimal, dataset-specific configuration that minimizes the overall dataset size. The final optimized weight matrix is then stored and subsequently evaluated alongside the two handcrafted predictors to assess performance improvements. The implementation is publicly available at https://github.com/xavifeme00/Astronomy-Compressor.

4. Experimental Results and Discussion

This section presents a comprehensive evaluation of the proposed compression approach and compares its performance against state-of-the-art lossless image compression methods commonly used in astronomical data processing. All results reported in this section are obtained using the test dataset for both the proposed method and the state-of-the-art techniques.

In our previous work (Maireles-González et al. 2023), the best-performing compressor for the INT, JKT, LCO, TJO, and WHT datasets was identified as JPEG 2000 using the Haar wavelet transform (J2K Haar). At that time, J2K Haar consistently outperformed the compression methods available within the FITS standard, as well as other widely adopted alternatives. However, for the newly introduced datasets—originating from more recent astronomical instruments—J2K Haar is no longer the best-performing compressor. In these cases, the strongest existing methods in the literature are JPEG-LS and JPEG XL, which achieve competitive compression ratios due to their advanced prediction schemes and efficient entropy coding mechanisms.

Table 2 summarizes the compression results obtained in this work and demonstrates a substantial improvement over previous findings. Across all datasets, even the most basic configuration of the proposed approach—the 2×2 predictor relying solely on the three nearest causal neighbors—already outperforms the best existing compressor for each dataset. This result highlights the effectiveness of the proposed predictor and entropy coding design. On average, this configuration achieves an improvement of approximately 0.10 bits per pixel (bpp) over the best competing compressor in macro average, which is J2K Haar. These results indicate that even a limited spatial context, when properly exploited and encoded, can yield significant compression gains. Further improvements are obtained by extending the spatial context of the predictor. The configuration employing the initial 9×9 causal neighborhood with uniformly initialized weights consistently outperforms the three-neighbor predictor across all datasets, confirming the benefits of exploiting broader spatial dependencies. For all datasets, this configuration yields the second-best overall compression performance. By expanding the spatial context without imposing differentiated weights within the 9×9 matrix, compression efficiency improves significantly. Compared to the 2×2 configuration, this approach achieves a further average improvement of 0.082 bpp, reaching a maximum gain of 0.141 bpp for the TJO and VLT datasets. This indicates that for this type of image, a smoother predictor is more robust than one highly sensitive to noise. When compared against the macro-average performance of J2K Haar, this configuration provides an average improvement of 0.177 bpp.

The best results across all datasets are produced by the third predictor, in which the weights are optimized. Compared with

Table 2

Compression Results in Bits Per Pixel (bpp) for the Three Proposed Predictors Compared with State-of-the-art Compressors

		DWT				Predictor							Dictionary		Others	
						Proposed Approach										
Telescope	**Entropy**	**J2K Haar**	**J2K 5/3**	**FAPEC 9/7**	**FPACK Hcompress**	**2 × 2**	**9 × 9**	**Optimized**	**JPEG XL**	**JPEG LS**	**CCSDS 123.0-B-2**	**HEVC**	**LZMA**	**Zstandard**	**bzip2**	**FPACK Rice**
INT	5.658	5.374	5.390	5.479	5.727	*5.342*	5.290	**5.266**	5.447	5.426	5.485	5.503	5.634	6.053	5.499	5.900
JKT	7.155	5.577	5.594	5.671	5.921	5.535	5.471	**5.447**	5.626	5.633	5.681	5.744	6.198	6.866	5.874	6.068
LCO	5.777	5.748	5.782	5.837	6.014	*5.654*	5.575	**5.566**	5.859	5.773	5.755	5.844	5.948	6.398	5.864	6.157
TJO	6.363	5.819	5.852	5.884	6.034	*5.752*	5.611	**5.611**	5.964	5.900	5.827	5.999	6.163	6.661	6.000	6.243
WHT	7.997	*6.402*	6.425	6.498	6.812	6.465	6.391	**6.363**	6.513	6.482	6.514	6.908	7.092	7.792	6.671	6.902
GTC	9.970	7.961	7.957	7.988	8.210	*7.843*	7.816	**7.755**	7.944	7.900	7.960	8.828	8.883	9.885	8.308	8.324
VLT	12.832	12.705	12.791	12.611	12.774	*12.335*	12.194	**12.187**	12.420	12.816	12.430	16.024	13.304	14.152	12.997	12.815
Micro Average	6.992	6.324	6.351	6.399	6.613	*6.253*	6.175	**6.156**	6.394	6.368	6.358	6.738	6.742	7.314	6.517	6.757
Macro Average	7.965	7.084	7.113	7.138	7.356	*6.989*	6.907	**6.885**	7.110	7.133	7.093	7.836	7.603	8.258	7.316	7.487

Note. The best result in each row is highlighted in **bold**, the second-best is underlined, and the third-best is shown in *italic*. The Micro Average accounts for the relative size of each dataset when computing the mean, whereas the Macro Average is obtained by averaging equally over all datasets.

the first and second handcrafted predictors, the optimized predictor provides additional gains of 0.104 bpp and 0.022 bpp on macro average. The maximum improvements are 0.148 bpp relative to the 2 × 2 predictor for the VLT dataset, and 0.061 bpp relative to the 9 × 9 predictor for the GTC dataset. Compared with the best state-of-the-art compressor, the optimized predictor achieves a substantial macro average improvement of 0.200 bpp across all datasets. These results show that training the weights specifically for each dataset, rather than using fixed weights, leads to a substantial improvement in compression performance, a factor that should be taken into account when designing the data storage pipeline in astronomical observatories.

A comparison with the best-performing FITS-standard compressor, FPACK Hcompress, further highlights the effectiveness of the proposed approach. The simplest 2 × 2 predictor already achieves an average improvement of 0.367 bpp (5.11%). Extending the spatial context to a 9 × 9 neighborhood increases this gain to 0.449 bpp (6.22%), while the optimized predictor delivers the best performance, with an average improvement of 0.471 bpp (6.40%). The maximum gains are consistently observed for the VLT dataset, peaking at 0.439, 0.580, and 0.587 bpp, respectively. Considering the VLT sensor's 2025 capability of capturing 7 TB of data per day, using FPACK Hcompress would compress this volume to approximately 5.58 TB. In contrast, the optimized predictor reduces the daily data volume to 5.33 TB, saving 0.25 TB per day. Over the course of a year, this results in roughly 91.25 TB of storage saved.

The observed improvements can be explained by the spatial characteristics of ground-based astronomical images. These images typically exhibit strong spatial correlations that extend beyond the immediate neighborhood of each pixel due to the optical point spread function, atmospheric effects, and the smooth background structure of astronomical scenes. As a result, predicting a pixel value using a broader spatial context can provide a more accurate estimate than predictors limited to only a few adjacent neighbors. In this work, the proposed predictor is able to exploit a substantially larger causal neighborhood than many traditional predictors used in astronomical compression algorithms. While classical predictors, such as those employed in standard compressors or in the spatial predictor of the CCSDS 123.0-B-2 recommendation, rely on only a few nearby pixels, the proposed method can incorporate information from a wider region around the current pixel. This extended spatial context enables the predictor to better capture the statistical structure of astronomical images and therefore produce residuals with lower entropy. Furthermore, the optimization of the predictor weights for each dataset allows the model to adapt to the specific statistical properties of images produced by different instruments. Since the distribution of pixel intensities, noise characteristics, and spatial correlations may vary between sensors, learning dataset-specific weights provides additional compression gains compared with fixed handcrafted predictors. Additionally, the proposed entropy coder further exploits contextual correlations. These results suggest that dataset-adapted predictors can play an important role in improving compression efficiency for modern astronomical data pipelines.

5. Conclusions

The volume of astronomical data generated by ground-based observatories is growing exponentially. Efficient storage management is an immediate necessity, but the FITS standard remains dependent on compression methods that do not fully exploit the capabilities of modern coding techniques. This study demonstrates that the use of linear predictors in combination with modern entropy coding techniques consistently outperforms the built-in compression methods of the FITS standard, such as FPACK Hcompress. Our experimental results also show that extending the spatial context and optimizing predictor weights through adaptive learning achieves an average compression efficiency improvement of 6.40% over FPACK Hcompress, which is the best-performing compressor currently available within the FITS framework.

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